

AYUSH GUPTA

kathilshiva@gmail.com | +91-7007226872

[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Leetcode](#) | [CodeChef](#) | [Codeforces](#)

EDUCATION

Vellore Institute of Technology, Bhopal

Bachelors of Technology in Computer Science and Engineering

CGPA: 7.96/10

Bhopal, India

2024 - 2028

Relevant Coursework: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Operating Systems

EXPERIENCE

Open-Source Contributor | Kubeflow Pipelines (Google / Linux Foundation)

Mar 2026 - Present

- Patched a critical Denial of Service (DoS) vulnerability in the Go backend by enforcing strict byte limits on artifact uploads to prevent zip bomb crashes [PR [#14171](#)].
- Secured the metrics parsing engine against unbounded memory exploits by implementing precise traversal budgets, eliminating Out-of-Memory (OOM) attacks [PR [#14186](#)].

Open-Source Contributor | DevPath

May 2026 - Present

- Fixed a major bug causing 500 Internal Server Errors by adding safe JSON data parsing, ensuring the backend doesn't crash when handling broken links [PR [#765](#)].
- Improved the recommendation algorithm using partial string matching for better search accuracy, expanded the database with 15 new projects, and fixed failing CI/CD tests to keep the codebase stable [PR [#804](#)].

PROJECTS.

System and Method of Predictive Safe Route Navigation Engine | ML, Next.js Python, Scikit-Learn, Leaflet, Geospatial APIs

[GitHub](#)

- Developed predictive navigation app for women's safety (Next.js map, FastAPI Python backend).
- Trained and deployed ML model (**Scikit-learn**) on crime datasets to dynamically calculate route safety scores.
- Designed low-latency spatial UI flagging high-risk zones and computing safest walking paths.

Smart File Organizer | Java, Swing, Java NIO, SHA-256, File I/O, Multithreading

[GitHub](#)

- Developed a Java desktop application that automatically organizes thousands of files using recursive scanning and configurable classification rules.
- Implemented SHA-256 duplicate detection, multithreading, and undo support for safe and efficient file management.
- Developed a hybrid CLI/Swing interface with file preview, progress tracking, and configurable sorting for user experience.

Retrieval-Augmented Generation Chatbot | Python, LangChain, ChromaDB, FAISS

[GitHub](#)

- Built a Retrieval-Augmented Generation system using LangChain, FAISS, and ChromaDB with sub-300 ms retrieval latency.
- Developed automated document ingestion pipeline (pdfjs) for text extraction, AI embedding generation, and fast retrieval.
- Designed interactive chat with Smart Command Palette and real-time Artifacts Canvas for intuitive user interactions.

POSITION OF RESPONSIBILITY

Core Member, Technical Team | MATRIX Club, VIT Bhopal

Aug 2024

- Collaborated with student leaders to foster a culture of open-source contribution, problem-solving, and technical excellence.

SKILLS

Programming Languages:	Python, Go, Java, C++, JavaScript
Machine Learning:	Deep Learning, Open CV, NLP, NLU, GenAI, RAG, LLMs, Information Retrieval
Frameworks & Libraries:	PyTorch, Scikit-learn, LangChain, spaCy, Ollama, FastAPI, Next.js
Databases:	PostgreSQL, MongoDB, ChromaDB, FAISS, Supabase
Software Engineering:	DSA, OOP, REST APIs, Distributed Systems, Git

ACHIEVEMENTS

- Selected as a Software Engineering Intern at **HackerRank**.
- Codeforces, Global Rank 166 among 30,000+ participants.
- Open-source contributor to **Kubeflow** Agent, Pipelines and Katib under Google and the Linux Foundation.
- Published** Indian Patent Application "System and Method of Data Accessibility in a Storage Device" (No. 202621077168).
- Earned certifications in **Applied Machine Learning** (University of Michigan) and **Generative AI** (Google Cloud).
- Successfully completed the **Deloitte** and **ServiceNow** Virtual Internship.